

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIOS & IDENTITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes	8 PYQs	Quantitative Analysis
Moderate	Trigonometric Ratio And Identites — Core Principles & Governing Laws	4 PYQs	Core Principle
Standard	Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes	$\tan^4 A + \cot^4 A =$, $\quad 2\sin A \sin B = \cos(A-B) - \cos(A+B)$, $\quad \tan A + \cot A = 2$
Trigonometric Ratio And Identites — Core Principles & Governing Laws	$\frac{\pi}{32}$, $\quad \sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}}$, $\quad y = \sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}}$
Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps	$\cos 5\pi = \cos(\pi + 4\pi) = -1$, $\cos 4\pi = 1$, $\quad \sin 2\pi = 0$, $\sin(-2\pi) = -\sin 2\pi = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes (High Yield — 8 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The maximum value of $\sin(x+(\pi/6))+\cos(x+(\pi/6))$ is attained at $x=...$ " [Ans: D]
- [2025 Shift] Q#4 (MCQ): "The minimum value of $1-\sin x$ is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The value of $\sin(5\pi/12)\sin(\pi/12)$ is..." [Ans: N/A]

3.2 Trigonometric Ratio And Identities — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Ratio And Identities.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Ratio And Identities PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If α and β are acute angles such that $\alpha-\beta$ and $\alpha+\beta$ satisfy the equation $\tan^2\theta-4\tan\theta+1=0$, $t...$ " [Ans: A]
- [2022 Shift] Q#1 (MCQ): "The degree measure of $(\pi/32)$ is equal to..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): " $\sqrt{(2+\sqrt{2}+\sqrt{2}+2\cos 8\theta)}=...$ " [Ans: N/A]

3.3 Trigonometric Ratio And Identities — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Which of the following is not correct?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, then α and β are respectively,

- (A) $45^\circ, 30^\circ$ [Correct]
- (B) $30^\circ, 45^\circ$
- (C) $30^\circ, 60^\circ$
- (D) $60^\circ, 45^\circ$

Correct Answer: A

Solution Analysis: Put $x = \tan\theta$. The given equation is $\tan^2\theta - 4\tan\theta + 1 = 0$ So, $x^2 - 4x + 1 = 0$ Using the quadratic formula, $x = \frac{4 \pm \sqrt{(16-4)}}{2}$ $x = \frac{4 \pm 2\sqrt{3}}{2}$ $x = 2 \pm \sqrt{3}$ Now, $\tan 15^\circ = 2 - \sqrt{3}$ and $\tan 75^\circ = 2 + \sqrt{3}$ So the two angles satisfying the equation are 15° and 75° Here the angles are $\alpha - \beta$ and $\alpha + \beta$. Since $\alpha + \beta$ is greater than $\alpha - \beta$, $\alpha - \beta = 15^\circ$ and $\alpha + \beta = 75^\circ$ Adding both equations, $2\alpha = 90^\circ$ $\alpha = 45^\circ$ Now, $\alpha - \beta = 15^\circ$

Benchmark Problem 2: KCET 2026 (Q#2)

The maximum value of $\sin(x + (\pi/6)) + \cos(x + (\pi/6))$ is attained at $x =$

- (A) $(\pi/2)$
- (B) $(\pi/4)$
- (C) $(\pi/6)$
- (D) $(\pi/12)$ [Correct]

Correct Answer: D

Solution Analysis: Let $y = x + (\pi/6)$ Then the expression becomes $\sin y + \cos y$ We know that $\sin y + \cos y = \sqrt{2} \sin(y + (\pi/4))$ The maximum value of $\sin(y + (\pi/4))$ is 1. So the maximum value is attained when $y + (\pi/4) = (\pi/2)$ This gives $y = (\pi/4)$ Now substitute back: $x + (\pi/6) = (\pi/4)$ $x = (\pi/4) - (\pi/6)$ $x = (3\pi - 2\pi)/12$ $x = (\pi/12)$ So the correct option is $(\pi/12)$ Option D.

Benchmark Problem 3: KCET 2025 (Q#3)

Which of the following is not correct?

- (A) $\cos 5\pi = \cos 4\pi$
- (B) $\sin 2\pi = \sin(-2\pi)$
- (C) $\sin 4\pi = \sin 6\pi$
- (D) $\tan 45^\circ = \tan(-315^\circ)$

Correct Answer: None

Solution Analysis: Let's check each identity: Option A: $\cos 5\pi = \cos(\pi + 4\pi) = -1$, $\cos 4\pi = 1$ These are not equal, so A is false. Option B: $\sin 2\pi = 0$, $\sin(-2\pi) = -\sin 2\pi = 0$ Option C: $\sin 4\pi = 0$, $\sin 6\pi = 0$ Option D: $\tan 45^\circ = 1$, $\tan(-315^\circ) = \tan(45^\circ) = 1$ Only Option A fails, so the incorrect statement is Option A.